

Practice Quiz: Cognition — The Mental Models

Running Your Life

Part A: Multiple Choice

1. A **mental model** is best described as: A) A plastic model you build from a kit B) Your brain's simplified version of how something works C) A perfect copy of reality D) Something only scientists have

Answer: B — Mental models are simplified maps your brain uses to navigate situations without analyzing everything from scratch.

1. Why does your brain build mental models? A) To waste energy B) To save you from analyzing every situation from zero C) To make you confused D) To eliminate all uncertainty

Answer: B — Models are "cheat sheets" that help you navigate familiar situations quickly and efficiently.

1. You act differently around friends versus grandparents because: A) You're being fake B) Your brain runs different mental models for different situations C) You have multiple personalities D) Your grandparents force you to change

Answer: B — Different contexts trigger different models for how to behave. It's natural adaptation, not fakeness.

1. Your mental model from 5th grade says "presentations = humiliation." In 7th grade, this model is: A) Still perfectly accurate B) Probably outdated and giving you glitchy results C) Something you should never question D) Proof you'll always hate presentations

Answer: B — Models built on old experiences can become outdated as you and your situation change.

1. The best way to fix an outdated mental model is to: A) Ignore it and hope it goes away
B) Test it by doing the thing and letting new evidence update it C) Avoid the situation forever D) Complain about it to everyone

Answer: B — Testing your model with new experiences lets your brain update it with fresh evidence.

1. **Metacognition** means: A) Memorizing more facts B) Thinking about your own thinking
C) Thinking faster than everyone else D) Never making mistakes

Answer: B — Metacognition is the ability to observe and question your own thought processes.

1. Which of these is a metacognition move? A) "I'm not thinking about anything" B) "I notice I'm making an assumption right now — what evidence do I have?" C) "I don't need to think about this" D) "I always trust my first instinct no matter what"

Answer: B — Naming your assumption and questioning the evidence is classic metacognition.

1. Mental models can go wrong because they're built from: A) Perfect information B) Limited experience that might be biased or outdated C) Universal truth D) Other people's brains

Answer: B — Models are based on your personal experience, which means they can be biased or no longer accurate.

1. A GPS with outdated maps is like: A) A brain that's always right B) A mental model built on old information that steers you wrong C) A perfect navigation system D) A brain that has no models at all

Answer: B — Outdated maps and outdated mental models both send you in the wrong direction.

1. Students who use metacognition tend to: A) Study less effectively B) Overthink everything and freeze up C) Study more effectively, manage stress better, and make smarter social decisions D) Avoid all challenges

Answer: C — Metacognition helps you step back, check your thinking, and make better choices.

Part B: Short Answer

1. Pick one mental model you have about school (like "math is hard" or "group projects are a waste of time"). Where did this model come from? Is it still accurate?
2. Describe a situation where your mental model about a person turned out to be wrong. What happened that updated your model?
3. How could you use metacognition the next time you're stressed about a test? Walk through the three moves: name it, question it, test it.
4. Think of a mental model that worked well for you in elementary school but doesn't quite fit in middle school. What changed, and what would a better model look like?
5. How could understanding mental models help you be a better friend to someone who sees a situation completely differently than you do?